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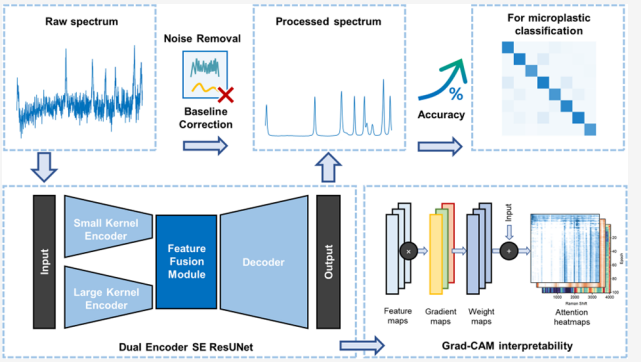
Enhancing Microplastic Raman Spectral Quality and Classification via a Dual Encoder Residual Network

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ABSTRACT: When addressing noise interference and baseline drift during the rapid Raman spectral analysis of microplastics, traditional denoising and baseline correction algorithms frequently exhibit limitations such as parameter sensitivity, reliance on manual intervention, and inadequate capacity for complex signal processing. In this work, a Dual Encoder ResUNet combined with attention mechanisms is proposed. Parallel encoders with small and large convolution kernels are adopted to achieve the extraction of multiscale spectral features. Furthermore, the internal decision-making mechanism of the model is investigated using Grad-CAM visualization technology, and the rationality of the dual-branch architecture is supported. Evaluated on a data set comprising 25,000 simulated training spectra alongside an independent testing set, the proposed model achieved an over 24% improvement in key metrics, including SNR, PSNR, and SSIM, compared to traditional Savitzky−Golay (SG) filtering and Wavelet Threshold Denoising (WTD). When applied to 19,200 Raman spectra of varying quality from eight types of microplastics collected under nonideal experimental conditions characterized by insufficient laser power and short acquisition times, the method enhanced the SNR by 3−15 times relative to traditional approaches. In downstream classification tasks, a classification accuracy of 82.80% is achieved by the Raman spectra processed by this model under the most challenging testing conditions, which is a substantial improvement over the 66.83% achieved by traditional methods and significantly higher than the 41.54% obtained with unprocessed data. These results demonstrate the advantages of the proposed neural network-based Raman spectral processing method in improving spectral quality and enhancing postprocessing performance.



INTRODUCTION

Microplastics are recognized as globally ubiquitous pollutants. Aquatic and terrestrial ecosystems are threatened by physical damage and chemical toxicity, while potential health risks are posed by the accumulation of microplastics in multiple human organs. Consequently, an urgent demand for efficient and accurate analytical tools is driven by these multifaceted ecological hazards and the compositional complexity of microplastics.[1−3] Currently, various detection methods are employed, including microscopy, infrared spectroscopy, Raman spectroscopy, fluorescence probes, mass spectrometry, thermogravimetric analysis, and liquid chromatography−mass spectrometry.[4−7] Among these, Raman spectroscopy is widely utilized as an optical technique for qualitative analysis due to its nondestructive nature, high specificity, and sensitivity.[8−12]

However, in real-time rapid detection scenarios within complex environments, significant disparities are observed between practically acquired Raman spectra and those obtained in laboratory settings. These disparities, including reduced signal-to-noise ratios, attenuated Raman peak intensities, decreased peak separation, and baseline drifts, present challenges to the classification accuracy of neural networks. Traditional baseline correction and denoising methods, along with their derivative algorithms, have been extensively investigated for Raman spectral data processing.[13,14] Specifically, penalized least-squares[15] and polynomial fitting[16] are widely employed for baseline correction. For denoising, techniques encompass Savitzky−Golay (SG) filtering,[17] Wavelet Threshold Denoising (WTD),[18] Whittaker Smoothing (WS),[19] Singular Value Decomposition (SVD),[20] and the Hilbert−Huang Transform (HHT).[21] These conventional methods have been widely applied to enhance the precision and reliability of signal analysis in Raman spectroscopy and other analytical instrumentation.[22] Despite their extensive utilization, traditional approaches are inherently associated with limitations, including

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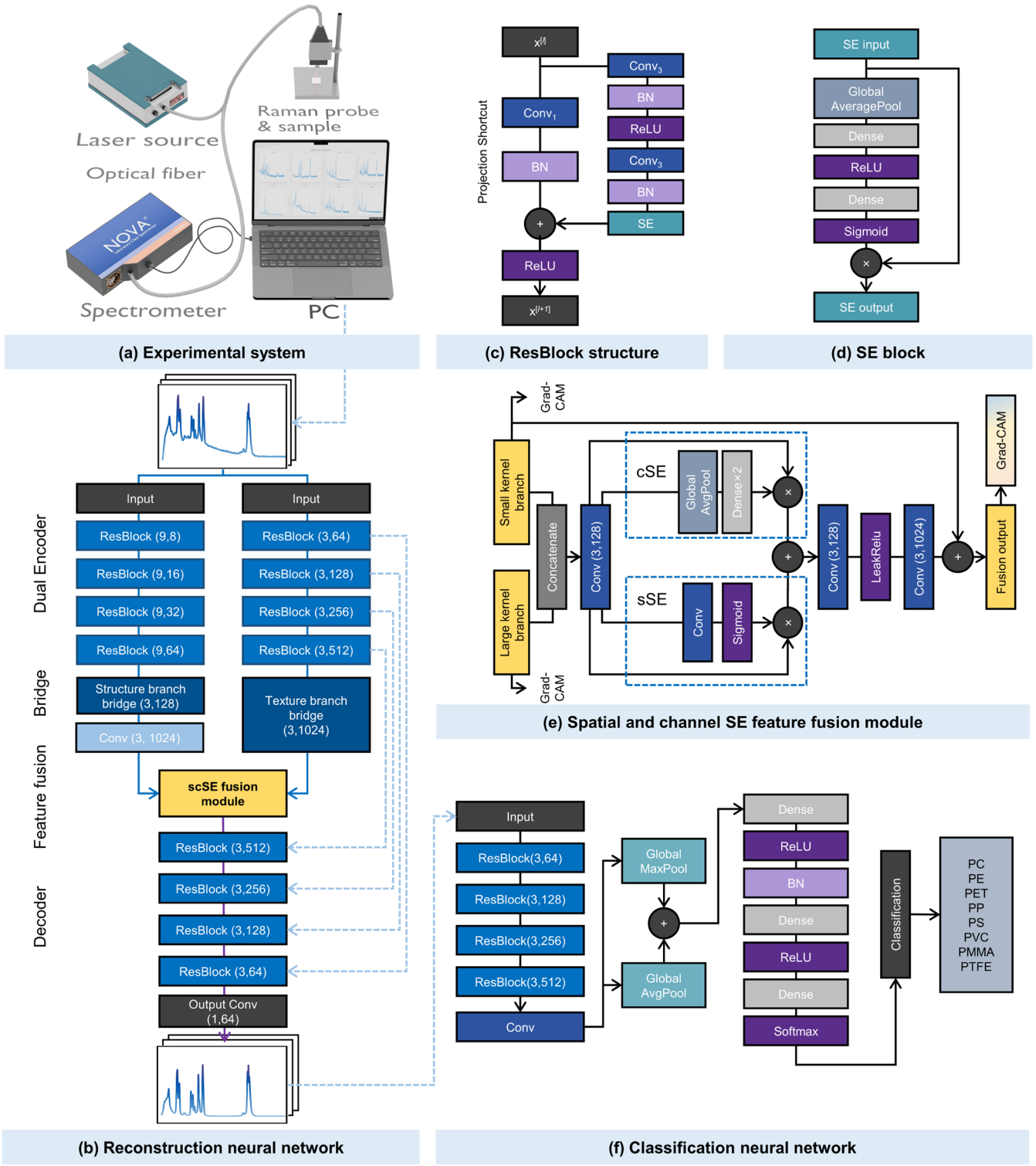


Figure 1. (a) Experimental system. (b) Neural network architecture for spectral denoising and baseline correction. (c) Residual block structure. (d) SE attention block structure. (e) Spatial and channel SE (scSE) feature fusion module. (f) Classification neural network structure.

high parameter sensitivity, poor generalization ability, and low computational efficiency in the processing of voluminous data sets.

To address these challenges, neural networks offer distinct advantages in efficiency, accuracy, and automation, enabling rapid high-throughput analysis. Neural network algorithms have emerged as a pivotal frontier for enhancing the quality of Raman spectra.[23] Within this framework, U-Net and analogous encoder-decoder architectures, which were originally developed for biomedical image segmentation, have become predominant due to their exceptional feature reconstruction capabilities.[24−26] Numerous studies have been conducted to further refine the feature extraction capability of these models. To mitigate information loss resulting from the fixed kernel sizes of traditional convolutions, the U-Net downsampling module was modified by Zeng et al.[27] through the integration of

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multiscale convolutions. An attention-based U-Net was proposed by Yang et al.[28] to exploit the intrinsic correlations between high and low frequency intramolecular vibrational modes, allowing low frequency features to be reconstructed from high frequency data. Furthermore, Transformer and BiLSTM modules were introduced by Zhao et al.[29] and Xue et al.,[30] respectively, to enhance the capacity of the model to capture global long-range dependencies within spectra. These studies have effectively enhanced feature representation through the integration of attention mechanisms or specialized modules within a single encoder pathway. Diverging from this approach, the dual encoder architecture, characterized by its parallel feature extraction, has demonstrated superior efficacy in capturing both local fine grained textures and global contextual information within the field of image processing.[31,32]

To ensure precise feature extraction and reconstruction within complex noise environments, an asymmetric dual encoder residual network integrated with attention mechanisms, designated as Dual Encoder Squeeze-and-Excitation ResUNet (DESE-ResUNet), is proposed in this work. The dual encoder is designed to extract multiscale features of detail and structural information, aiming to enhance its compatibility with Raman spectral preprocessing tasks, which require simultaneous preservation of characteristic peaks and removal of baselines. Squeeze and Excitation (SE) blocks are integrated into the network, and a Spatial and Channel SE (scSE) dual branch feature fusion module is introduced in the bridging layer to facilitate the effective integration of dual stream features across spatial and channel dimensions. Furthermore, to investigate the internal decision-making basis of the model, gradient-weighted class activation mapping (Grad-CAM)[33−35] is employed to visualize the evolution of important features during the training process, aiming to clarify the collaborative mechanism of the dual branch architecture in spectral denoising and baseline correction. The proposed method achieves effective denoising and baseline correction for Raman spectra. Compared with conventional algorithms, it demonstrates significant advantages in overall performance, including improved signal-to-noise ratio enhancement, effective fluctuation suppression, and enhanced post processing efficacy across varying noise levels.

METHODS

Raman Detection

The Raman spectroscopic system used in this work is consistent with the system employed in our previous work.[36−38] Its core configuration primarily comprises a laser source, a Raman probe, a Raman spectrometer, and a computer. To balance fluorescence effects and excitation efficiency, a 785 nm laser (Changchun New Industries Optoelectronics Technology Co., Ltd.; FCD-785) was selected as the excitation source. This laser delivers a maximum output power of 515 mW with a line width of 0.06 nm. The Raman probe (RPB-D-A, Changchun New Industries Optoelectronics Technology) was coupled to a laser with a central wavelength of 785.442 nm, achieving a coupling efficiency exceeding 50%. Spectral acquisition was performed using a Raman spectrometer (Shanghai Ideaoptics Corporation; NOVA2S) with a resolution of 1.04 nm and Raman shift range spanning from 132 cm^{-1} to 4052 cm^{-1}.

Eight representative polymer types were selected for analysis: polycarbonate (PC), polyethylene (PE), polyethylene terephthalate (PET), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC), poly(methyl methacrylate) (PMMA), and polytetrafluoroethylene (PTFE). The samples encompassed six distinct particle sizes: 10, 44, 100, 350, 2000, and 4000 μm (PE, PP, PS, and PVC from Ruixiang Polymer Materials, China; PC, PMMA, and PTFE from Jiecheng Plasticizing Co., China). The particle size of microplastic samples refers to the average particle size provided by the manufacturer. Actual powdered microplastic samples exhibit irregular particle sizes. For each particle size fraction, 50 Raman spectra were acquired under varying laser power and integration time settings, with each spectrum consisting of 1024 distinct spectral points.

Neural Network Structure

In contrast to the traditional U-Net, ResUNet incorporates residual blocks (Figure 1c) utilizing projection shortcuts to facilitate direct gradient flow through network layers. This mechanism effectively mitigates the vanishing or exploding gradient problems inherent in deep neural network training, ensuring continuous optimization as network depth increases.[39] To better balance the preservation of weak signals and the removal of baselines and noise, with the objective of enhancing the neural network's capacity to extract multiscale information features in complex noise, this work employs an asymmetric dual encoder network (Figure 1b). In contrast to the single-encoder models utilized in previous studies,[37,38] this design leverages two parallel encoder branches to extract features across different scales, thereby providing the decoder with a richer global feature representation. The Texture Encoder features a high initial channel count (64) and a small kernel size (3). This configuration adopts the classic design of the U-Net architecture, where small convolution kernels effectively capture local variations in peak morphologies while maintaining low parameter counts and computational costs. Alongside this efficiency, the larger initial channel count provides the network with ample feature representation capacity. Following four stages of downsampling, the output feature channel count of this branch expands to 1024, thereby enabling the comprehensive characterization of the intricate and rich peak details within the Raman spectra. Conversely, the Structure Encoder employs a reduced initial channel count (8) paired with a large kernel size (9). Rather than pursuing abundant local texture representations, this branch aims to capture the overall spectral trends, baseline drifts, and broad peak structures within a larger receptive field. The deliberate adoption of a smaller initial channel count mitigates the parameter inflation typically associated with large convolution kernels, thereby preventing the structure branch from exerting an excessive impact on the overall model complexity. Through these structural choices, this specific encoder branch directs the network to focus on extracting the approximate shape features of the spectra. The parameter sensitivity analysis regarding kernel sizes and initial channel counts is presented in Table S3. Building upon this architecture, a Squeeze-and-Excitation (SE) block (Figure 1d)[40] was integrated into the residual block. The SE block operates by first employing a global average pooling squeeze operation to aggregate feature maps across spatial dimensions, thereby capturing the global distribution of channel descriptors. Subsequently, an excitation mechanism models channel wise dependencies. This is implemented via two fully connected layers equipped with ReLU and Sigmoid activation functions, respectively. Finally, a dual branch feature fusion module based on concurrent spatial and channel squeeze and excitation (scSE)[41] (Figure 1e) was integrated at the encoder-decoder bridge. This module fuses features from both encoder branches, equipping the decoder with richer and more discriminative global feature representations. The proposed model is designated as Dual Encoder SE-ResUNet (DESE-ResUNet). The complete model implementation and training code are publicly available at: https://github.com/V1S10NAL/DESE-ResUNet.

$\mathbf{X}_{sig} \in \mathcal{R}^{L\times C}$ and $\mathbf{X}_{noise} \in \mathcal{R}^{L\times C}$ denote the feature maps derived from the signal encoder and noise encoder at the bridge layer, respectively, where L represents the spectral length and C denotes the channel count (C = 1024 in this work). The fusion process employs an scSE mechanism to extract differential features, which are subsequently injected into the main branch via a residual connection. First, the two feature sets are concatenated along the channel dimension and compressed via a convolutional layer to an intermediate channel depth C' (C' = 128), to yield the fused feature representation $\mathbf{Z} \in \mathcal{R}^{L\times C}$:

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$$\mathbf{Z} = \phi(\mathcal{F}_1^{3\times1}([\mathbf{X}_{sig}, \mathbf{X}_{noise}])) \quad (1)$$

where $[\cdot, \cdot]$ denotes the concatenation operation, $\mathcal{F}_1^{3\times1}$ represents a 1D convolution with a kernel size of 3, $\phi(\cdot)$ represents the LeakyReLU activation function. The feature Z is then processed by the parallel scSE module. In the channel squeeze-and-excitation (cSE) branch, a global average pooling (GAP) operation followed by fully connected layers generates the channel weight vector $\mathbf{w}_c = \mathcal{R}^{1\times C}$:

$$\mathbf{w}_c = \sigma(\mathbf{W}_2\delta(\mathbf{W}_1 g(\mathbf{Z}))) \quad (2)$$

$$\mathbf{Z}_{cSE} = \mathbf{w}_c \odot \mathbf{Z} \quad (3)$$

where $g(\cdot)$ denotes the global average pooling operation, $\mathbf{W}_1 \in \mathcal{R}^{\frac{C}{2}\times C}$, and $\mathbf{W}_2 \in \mathcal{R}^{C,\times\frac{C}{2}}$ represent the weights of the fully connected layers, $\delta(\cdot)$ represents the ReLU function, $\sigma(\cdot)$ refers to the Sigmoid function, and $\odot$ represents element-wise multiplication. The spatial squeeze-and-excitation (sSE) branch via a 1×1 convolution generates a spectral-spatial mask $\mathbf{M}_s \in \mathcal{R}^{L\times1}$:

$$\mathbf{M}_s = \sigma(\mathcal{F}_s^{1\times1}(\mathbf{Z})) \quad (4)$$

$$\mathbf{Z}_{sSE} = \mathbf{M}_s \odot \mathbf{Z} \quad (5)$$

The fused attention feature $\mathbf{Z}_{att}$ is subsequently obtained as the element-wise sum of the two branches:

$$\mathbf{Z}_{att} = \mathbf{Z}_{cSE} + \mathbf{Z}_{sSE} \quad (6)$$

Finally, the features are restored to the original dimension C, via a feature reconstruction and projection layer. These are then added to the main signal branch as a residual component to yield the final fused output $\mathbf{Y} \in \mathcal{R}^{L\times C}$:

$$\mathbf{R} = \mathcal{F}_{proj}^{1\times1}(\phi(\mathcal{F}_{rec}^{3\times1}(\mathbf{Z}_{att}))) \quad (7)$$

$$\mathbf{Y} = \mathbf{X}_{sig} + \mathbf{R} \quad (8)$$

where $\mathbf{R}$ denotes the detailed residual derived from the noise branch.

To evaluate the impact of neural network-based denoising on downstream analysis, a ResNet model was employed for spectral classification. This experiment compares the effects of various data preprocessing strategies on classification performance (Figure 1f).

Simulated Data Generation

To address the substantial data requirements for neural network training and facilitate quantitative performance evaluation, simulated data sets were utilized in this study. Consequently, Raman spectral peaks are commonly approximated using the Voigt profile, which is obtained by the convolution of Gaussian and Lorentzian functions.[42] Due to the absence of an elementary analytical solution for this integral, numerical discrete integration was utilized to compute the convolution (eq 9). Each simulated spectrum, consisting of 1024 points, was constructed by superimposing n independent Voigt peaks. The continuous integral definition of the Voigt profile and its discrete approximation are expressed as follows:

$$V(x; \sigma, \gamma) = \int_{-\infty}^{+\infty} G(t, \sigma)\cdot L(x - t; \gamma)\mathrm{d}t \approx \sum_{k=1}^{M} G(t_k, \sigma)\cdot L(x_k - t_k; \gamma)\Delta t \quad (9)$$

$$G(t, \sigma) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\left(\frac{t^2}{2\sigma^2}\right)\right) \quad (10)$$

$$L(x - t, \gamma) = \frac{\gamma}{\pi(x - t - \mu)^2 - (\gamma)^2} \quad (11)$$

$$R(x) = \sum_{i=1}^{n} A_i\cdot V(x - \mu_i, \sigma_i, \gamma_i) \quad (12)$$

where the convolution kernels $G(t; \sigma)$ and $L(x-t; \gamma)$ represent the Gaussian and Lorentzian components, respectively. $R(x)$ denotes the pristine Raman spectrum generated by the linear superposition of n independent Voigt peaks. x represents the spectral wavenumber, σ is the standard deviation of the Gaussian component, and γ is the half-width of the Lorentzian component. M is the integration step count (M = 1024). For the i-th peak, μ_i represents the peak center position, A_i represents the amplitude. The value of A_i is adjusted to ensure that the intensity of the generated spectrum approximates that of normalized experimental spectra. The total number of Raman peaks, n is selected randomly from 5 to 30. The peak widths (in cm^{-1}) are randomly assigned, with 50% for primary peaks within the range of 19.2 to 55.2, 35% for secondary peaks within 30.6 to 69.0, and the remaining 15% for tertiary peaks within 61.3 to 459.7. To ensure physically reasonable peak shapes, the maximum amplitude of the secondary peaks is restricted to 50% of that of the largest primary peak, and the maximum amplitude of the tertiary peaks is limited to 20% of the largest primary peak. Finally, the sampling range for the Raman peaks aligns with the spectral range of the employed spectrometer, spanning from 132.57 to 4051.96 cm^{-1}.

To simulate realistic experimental environments, the final generated Raman spectrum is constructed as a superposition of the pristine spectrum $R(x)$, a polynomial baseline $B(x)$, and a noise component N:

$$R_{simulate} = R(x) + B(x) + N \quad (13)$$

$$B(x) = \sum_{j=0}^{K} c_j x^j,\ c_j = \arg\min_{\{c\}} \sum_{m=1}^{K+1} (y_m - B(x_m))^2 \quad (14)$$

where c represents the coefficients of the polynomial baseline $B(x)$, which are determined via least-squares fitting of randomly sampled points (x_m, y_m), and K denotes the order of the polynomial.

The noise component N is designed to encompass a broad range of practical measurement scenarios. To this end, three distinct noise generation mechanisms were established: (1) pure Gaussian noise; (2) a hybrid mixture comprising Gaussian, Poisson, and impulse noise; and (3) real environmental noise, acquired by the instrument in the absence of samples and subsequently standardized. The intensity of these noise components is regulated by the standard deviation of the baseline-augmented spectra. The specific definitions for these noise categories are as follows:

$$N_{Gaussian} \sim N(0, \sigma_n^2),\ \sigma_n = r\cdot\sigma_s \quad (15)$$

$$N_{Poisson} = r\cdot\left(\frac{\mathrm{Poisson}(a\cdot s_{shift})}{a} - s_{shift}\right) \quad (16)$$

$$N_{Impulse} = \mathrm{Bernoulli}(p)\cdot U(0.1F, F),\ F = r\cdot\max(s)\cdot b \quad (17)$$

$$N_{Measured} = r\cdot\sigma_s\cdot N_z,\ N_z = \frac{N_{raw} - \mu_{raw}}{\sigma_{raw}} \quad (18)$$

where $N_{Gaussian}$, $N_{Poisson}$, $N_{Impulse}$ and $N_{Measured}$ denote the Gaussian, Poisson, impulse, and measured noise components, respectively. σ_s represents the standard deviation of the baseline-augmented spectrum, r serves as the noise level coefficient, ranging from 0 to 100%. σ_n corresponds to the standard deviation of the added Gaussian noise. s_{shift} denotes the shifted non-negative baseline-augmented spectral signal, and a is the Poisson noise scaling factor (set to 0.01). Regarding impulse noise, p represents the probability of occurrence controlled by a Bernoulli distribution (set to 0.1), F denotes the maximum impulse amplitude, s is the baseline-augmented spectral signal (set to 0.2), and b is the spike amplitude ratio coefficient relative to s. N_z represents the standardized measured noise derived from the raw measured noise N_{raw}, with μ_{raw} and σ_{raw} denoting the mean and standard deviation of the raw noise, respectively.

Evaluation of Simulated Spectra Quality

The Natural Image Quality Evaluator (NIQE) serves as a widely adopted no-reference metric within the computer vision domain for

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Table 1. NSQE for Different Data Sets

	measured				simulated set 1			simulated set 2		
data set	group 1	group 2	group 3	group 4	hybrid noise	Gaussian noise	measured noise	hybrid noise	Gaussian noise	measured noise
NSQE	2.0595	2.0450	2.0584	2.0396	4.6414	4.2577	**4.2256**	4.8331	4.3700	**4.3382**

assessing the naturalness and distortion levels of noisy images.[43,44] Based on Natural Scene Statistics (NSS) theory, NIQE operates on the premise that high quality natural images, following specific normalization, exhibit pixel intensity distributions that adhere to specific statistical regularities, typically Gaussian or generalized Gaussian distributions. Distortions such as blur, noise, and artifacts disrupt these regularities. The construction of an ideal feature model derived from authentic natural images enables NIQE to quantify the naturalness of an image by computing the statistical distance between the features of the evaluated image and those of the model. Drawing inspiration from the application of NIQE in two-dimensional images, this work employs a one-dimensional adaptation of NIQE to evaluate the similarity between simulated and natural spectra. This is designated as the Natural Spectra Quality Evaluator (NSQE). Analogous to natural images, experimental measurements of spectral signals are constrained by physical laws and governed by intrinsic molecular vibrational properties and instrumental broadening effects.[45] The statistical distributions of their signal and noise features are not random but rather possess natural spectrum statistics characteristics.[25] The NSQE aims to assess the realism of simulated spectra by quantifying the distance between the statistical distributions of the target spectrum and authentic spectra. The calculation process of NSQE follows the core method of NIQE and has been adapted to the characteristics of one-dimensional spectral data. A lower NSQE score indicates that the statistical features of the evaluated spectrum are closer to the centroid of the authentic spectral distribution, thereby possessing higher realism. Conversely, a higher score indicates a greater deviation of the spectrum from the authentic spectra.

Specifically, NSQE first applies a 1D Mean Subtracted Contrast Normalization (MSCN) via a sliding window to eliminate global intensity fluctuations and baseline drift, thereby isolating the intrinsic peak profiles and noise textures. Subsequently, Generalized Gaussian Distribution (GGD) and Asymmetric Generalized Gaussian Distribution (AGGD) models are utilized to fit these normalized coefficients and extract a representative statistical feature vector. Ultimately, spectral naturalness is quantified by computing the single-point Mahalanobis distance between this extracted target feature vector and a reference Multivariate Gaussian (MVG) distribution constructed from authentic experimental spectra. Further details regarding NSQE are presented in Supporting Information Section S1.

Model Training

Utilizing the method described previously, a data set comprising 25,000 spectra was generated to train the spectral reconstruction neural network. This data set was partitioned into training, validation, and test subsets according to a 7:1:2 ratio. All computational experiments were implemented within the Python 3.8 and TensorFlow 2.10 framework, utilizing an NVIDIA GeForce RTX 4070 Ti SUPER (16 GB) GPU for acceleration. Detailed computational environment specifications and neural network hyperparameters are provided in Supporting Information Tables S1 and S2.

Evaluation Metrics

For the spectral denoising and baseline correction tasks, the Signal-to-Noise Ratio (SNR) was employed to quantify the signal quality of the simulated data sets,[46] while the Signal-to-Noise Ratio improvement (SNR*) served to measure the enhancement in spectral signal quality following the denoising process.[47] The Max-Referenced Fluctuation (MRF) was utilized to characterize the differences in spectral fluctuation across simulated noise levels. To assess the quality of the reconstructed spectral signals, the Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM),[42] and Cosine Similarity (CS)[25] were adopted as evaluation metrics. Furthermore, Receiver Operating Characteristic (ROC) curves and the corresponding Area Under the Curve (AUC) values[48] were used to evaluate classification performance across all thresholds. Detailed descriptions of these evaluation metrics are provided in Supporting Information Sections S1−S8.

Partial Least Squares Discriminant Analysis

In this work, we employed the classic chemometric method, Partial Least Squares Discriminant Analysis (PLS-DA), for comparison with the ResNet classification approach. PLS-DA is a linear supervised learning model that integrates the dimensionality reduction properties of partial least-squares with the discriminatory capabilities of classification algorithms.[49,50] Its core mechanism involves projecting the Raman spectral data as independent variables and the microplastic category labels as dependent variables into a lower-dimensional feature space. By maximizing the covariance between these variable sets, this approach effectively extracts latent variables that possess the greatest discriminatory power for differentiating among various microplastic categories. To evaluate the model comprehensively, the number of principal components was varied as the core hyperparameter, with experiments systematically conducted using 10, 15, and 25 principal components.

RESULTS AND DISCUSSION

NSQE Evaluation of Simulated Raman Spectra Data Sets

A total of 19,200 Raman spectra, acquired from eight types of microplastics under eight distinct experimental conditions, were partitioned into five equal groups at regular intervals. One group was selected to construct a standard multivariate Gaussian model to serve as the NSQE reference spectra. The NSQE values between the remaining four groups and the reference spectra were then calculated to provide a benchmark for evaluating the differences between the simulated data sets containing various types of noise and the true spectra. Two sets of pristine Raman spectra and polynomial baselines were generated using eqs 9−14. Building upon this foundation, three distinct simulated data sets were constructed according to eqs 15−18 to incorporate different noise types. These specific profiles consist of pure Gaussian noise, a hybrid mixture of Gaussian, Poisson, and impulse noises, as well as real environmental noise acquired experimentally in the absence of samples and individually standardized. Each individual data set comprises 25,000 spectra. The results presented in Table 1 demonstrate that the four groups of real-world spectra exhibit the lowest NSQE. Among the three types of simulated noise, the two sets of simulated spectra incorporating measured noise exhibit the lowest NSQE values. This indicates a minimal statistical discrepancy between these simulated spectra and the measured ones, suggesting that their spectral noise characteristics most closely resemble those of natural measured spectra. In subsequent experiments, these two simulated spectrum groups, with added real-world noise, will be used as the training data set and an extra test data set.

The DESE-ResUNet Neural Network Demonstrates Better Noise Removal and Baseline Correction Performance

The ResUNet, SE-ResUNet, CBAM-ResUNet, CSAM-ResUNet,[33] and DESE-ResUNet neural network models were trained using a simulated spectral data set comprising 25,000 spectra, with identical hyperparameter settings and random seed

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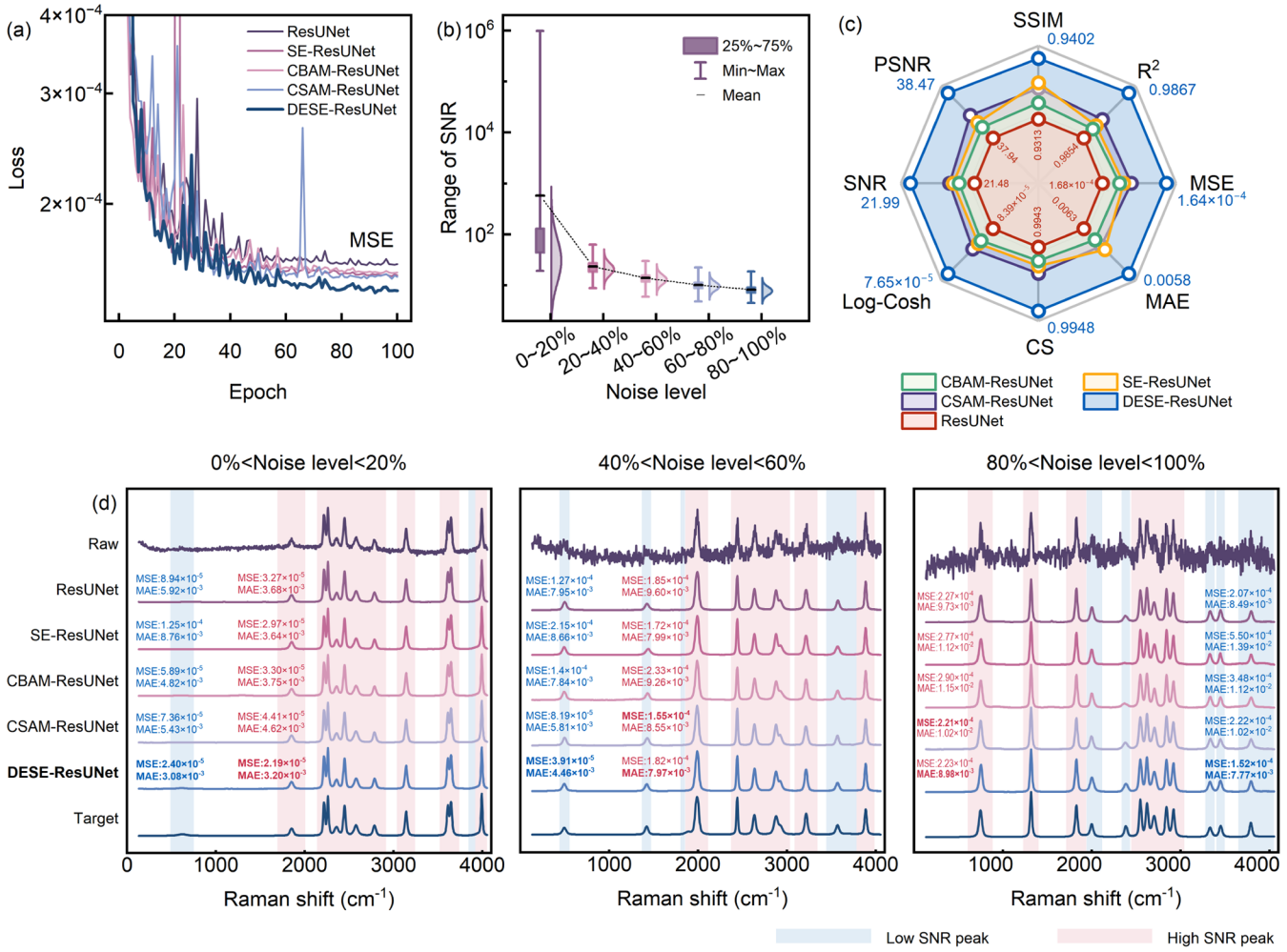


Figure 2. (a) Comparison of training loss curves. (b) SNR distribution of the extra test data set used to assess model performance at different noise levels. (c) Comparison of model performance across different metrics. (d) Comparison of noise removal and baseline correction effects among different neural networks at varying noise levels.

Table 2. Performance Comparison of Neural Network Models on the Extra Test Set

			full band range							
model	params/M	FLOPs	R^2	MSE	MAE	Log-Cosh	SNR*	PSNR	CS	SSIM
ResUNet	**12.94**	**3049.83**	0.9854	1.68×10^{-4}	6.29×10^{-3}	8.39×10^{-3}	21.45	37.94	0.9943	0.9313
SE-ResUNet	13.38	3051.88	0.9858	1.63×10^{-4}	6.08×10^{-3}	8.14×10^{-3}	21.63	38.12	0.9945	0.9366
CBAM-ResUNet	14.69	3057.63	0.9857	1.64×10^{-4}	6.18×10^{-3}	8.19×10^{-3}	21.58	38.07	0.9944	0.9337
CSAM-ResUNet	13.71	3128.65	0.9859	1.61×10^{-4}	6.12×10^{-3}	8.05×10^{-3}	21.66	38.21	0.9945	0.9358
DESE-ResUNet	14.66	3230.51	**0.9867**	$\mathbf{1.54 \times 10^{-4}}$	$\mathbf{5.83 \times 10^{-3}}$	$\mathbf{7.65 \times 10^{-3}}$	**21.99**	**38.47**	**0.9948**	**0.9402**

		high signal-to-noise ratio region			low signal-to-noise ratio region		
model	inference time/ms	R^2	MSE	MAE	R^2	MSE	MAE
ResUNet	$\mathbf{0.6987 \pm 0.0242}$	0.9900	2.27×10^{-4}	8.19×10^{-3}	0.9323	2.99×10^{-4}	9.43×10^{-3}
SE-ResUNet	0.7777 ± 0.0213	0.9902	2.23×10^{-4}	8.03×10^{-3}	0.9339	2.92×10^{-4}	9.30×10^{-3}
CBAM-ResUNet	0.9545 ± 0.0265	0.9902	2.24×10^{-4}	8.08×10^{-3}	0.9346	2.89×10^{-4}	9.22×10^{-3}
CSAM-ResUNet	0.9995 ± 0.0289	0.9904	2.19×10^{-4}	7.96×10^{-3}	0.9351	2.86×10^{-4}	9.09×10^{-3}
DESE-ResUNet	0.9776 ± 0.0210	**0.9908**	$\mathbf{2.09 \times 10^{-4}}$	$\mathbf{7.70 \times 10^{-3}}$	**0.9372**	$\mathbf{2.77 \times 10^{-4}}$	$\mathbf{9.08 \times 10^{-3}}$

configurations. Training the SE-ResUNet model took approximately 3 h. An extra data set of 25,000 spectra was generated to compare the performance of the models in terms of noise removal and baseline correction (Figure 2). During the training phase, following initial oscillations, the validation Mean Squared Error (MSE) loss of DESE-ResUNet decreases more rapidly in the later stages compared to the other models (Figure 2a). This improved convergence behavior suggests that DESE-ResUNet more effectively extracts the critical features necessary for spectral reconstruction, specifically denoising and baseline correction. In the extra data set, as the standard deviation of added noise increases, spectral noise interference intensifies and the signal-to-noise ratio decreases (Figure 2b), indicating that poorer spectral data quality poses a greater challenge to model

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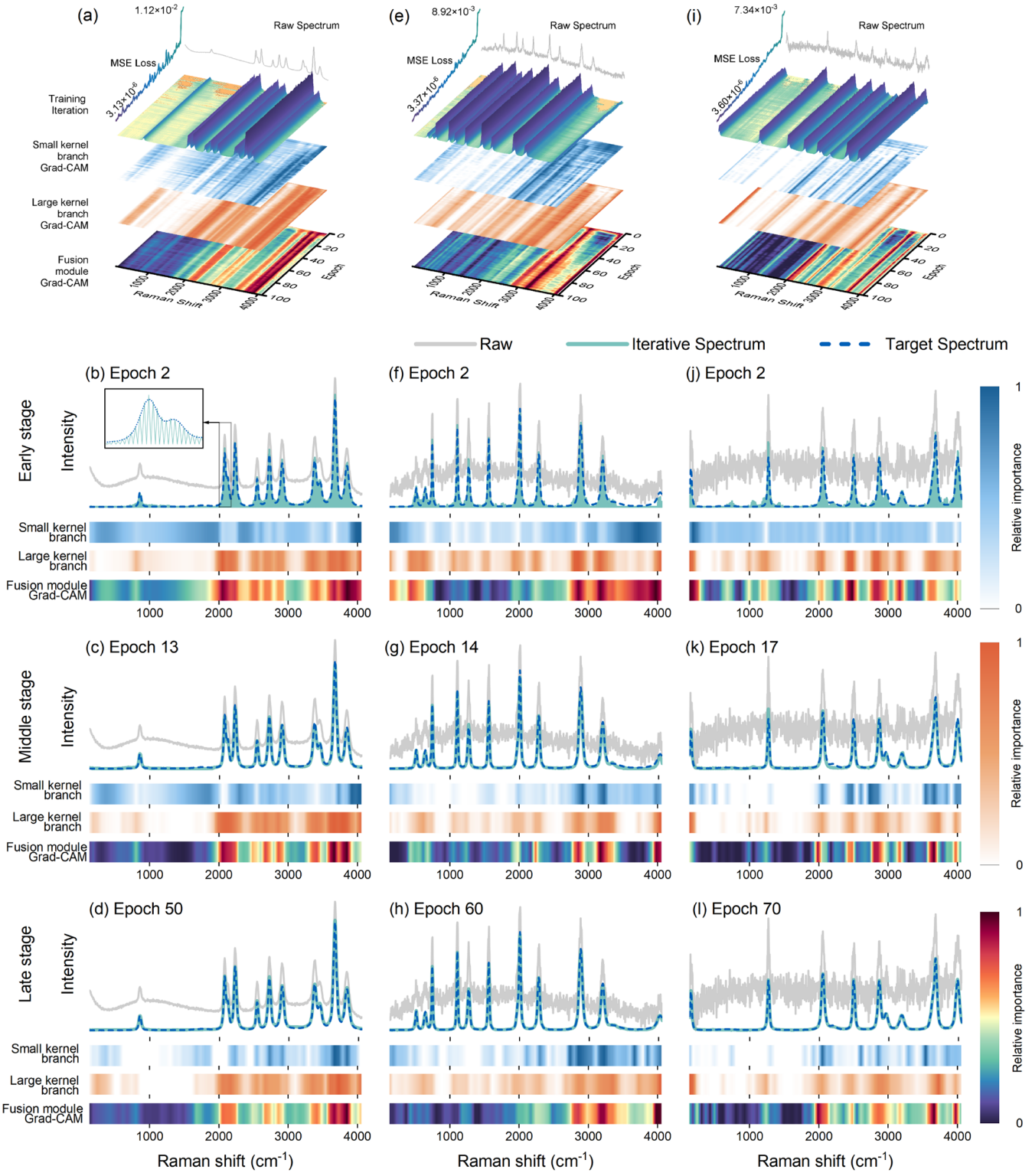


Figure 3. (a, e, i) Iterative spectral evolution of the training set during the model training process. Stacked Grad-CAM heatmaps visualizing feature importance were computed at the final convolutional layer of the small-kernel branch, large-kernel branch, and feature fusion module. The MSE loss represents the loss variation curve across iterations for this spectrum during model training. (b–d), (f–h), (j–l) Evolution of spectral iterations and feature importance across early, intermediate, and late training stages for spectra exhibiting low, medium, and high noise levels, respectively.

performance. In all spectral processing tasks on the extra test data set, the DESE-ResUNet model demonstrated superior performance compared to other models across multiple metrics. These include regression loss indicators such as R^2, Mean Squared Error (MSE), Mean Absolute Error (MAE), and Log-Cosh, as well as similarity metrics like CS and SSIM. The SE-ResUNet model enhanced the ResUNet MSE and MAE by 2.88% and 3.40%, respectively, while the DESE-ResUNet model achieved improvements of 8.76% and 7.35% (Table 2, Figure 2c).

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In order to make a more direct comparison of the performance of the models in terms of peak shape recovery, comparative analyses were conducted of regression performance in the peak region under different signal-to-noise conditions. The peak region was segmented into zones based on the relative ratio of noise amplitude to peak intensity. In instances where the noise amplitude within the peak region was less than 75% of the peak intensity, the area centered on the peak and spanning four times the line width was designated as a high signal-to-noise ratio region. Conversely, the area centered on the peak and spanning 12 times the line width was designated as a low SNR region (in order to ensure complete capture of the long-tail characteristics of low intensity Lorentzian or Gaussian peaks). In high SNR regions, DESE-ResUNet achieved improvements of 8.04% in MSE and 6.00% in MAE over ResUNet, whereas improvements of 7.21% and 3.76% were observed in low SNR regions. Conversely, the SE-ResUNet model yielded more modest gains of approximately 2−3%. Figure 2d illustrates the denoising performance across three representative spectra exhibiting low, medium, and high noise levels, with corresponding MSE and MAE values calculated specifically for high and low signal-to-noise ratio intervals. Collectively, the DESE-ResUNet model demonstrates superior proficiency in processing both low and high SNR peak regions. It exhibits enhanced capabilities in the identification and reconstruction of signals amidst significant noise interference (Table 2, Figure 2d).

To evaluate the trade-off between model accuracy and architectural complexity, Table 2 reports the parameter count, computational complexity (floating point operations), and actual inference time per spectrum for each model ablation variant. The inference times were determined using an additional test set of 25,000 spectra processed in batches, with the values expressed as the mean and standard deviation across five independent runs. As the feature extraction capability of the model is enhanced, its parameter count and theoretical computational complexity inevitably increase. Owing to the introduction of the secondary encoder, the parameter count and theoretical computational complexity of DESE-ResUNet increased by 9.57% and 5.85%, respectively, relative to the SE-ResUNet architecture. The inference time per spectrum increased by a mere 0.2 ms. Given that actual experimental spectral acquisition times generally range from hundreds to thousands of milliseconds, this marginal increase in processing time introduced by the additional architectural components is negligible. Consequently, it does not impede the high-speed, real-time processing efficacy of the model in practical applications.

Explainability Analysis of the Model Training Process

To investigate the decision basis of the dual-branch encoder during noise removal and baseline correction, this work employs Grad-CAM to visualize the importance of features in the model's internal feature extraction during the training process of the model. The intermediate spectral state was captured at each epoch during model training by utilizing a custom Keras callback function. Figure 3a, e, i illustrate the iterative evolution of the model's spectral outputs and the corresponding Grad-CAM importance feature trajectories across each epoch for three representative spectra from the training set, exhibiting low, medium, and high noise levels. To investigate the distinctions and correlations in feature extraction between the two branches, Grad-CAM visualizations were computed at the final convolutional layers of the dual encoder branches (large-kernel and small-kernel) immediately prior to the feature fusion module, as well as at the terminal convolutional layer within the fusion module itself. The color mappings for the small and large kernel branches utilize blue and orange gradients, respectively, while the fused features are visualized using the RedPurple color map from Origin. Regarding the normalization scheme, the feature contribution was computed independently for an individual input spectrum at each training epoch. A minimum to maximum normalization method was subsequently applied to dynamically map all activations into a relative importance scale ranging from 0 to 1. The flattened representations of the stacked heatmaps depicted in Figure 3a, e, i are presented in Figure S3. Visualizations illustrating the iterative evolution of more spectra during the model training process are presented in the Supporting Information Video.

The computed Grad-CAM heatmaps reveal distinct functional roles for the two parallel branches. During the early training stages (epochs 1−15; Figure 3b, f, j), the large-kernel encoder branch leverages the expansive receptive field provided by its large kernels to prioritize the coarse capture and approximate localization of core peak regions. Conversely, the small-kernel encoder branch focuses on extracting features from peak-free background regions, likely facilitating the restoration of a flat baseline. Notably, this operational state persists for a more extended duration in spectra with low noise levels (Figure 3c). This stage accomplishes the preliminary reconstruction of the coarse spectral contour. Remarkably, during the initial training process, the spectra are reconstructed as nonsmooth waveforms with sawtooth-like fluctuations, serving to approximate the general shape of the reference spectra (Figure 3b).

During the middle training stage (epochs 15−40), the spectral peak shapes become essentially established (Figure 3g, k). The large-kernel branch demonstrates a capacity to characterize the broad regions spanning the peak width, thereby ensuring the structural integrity of the peak profiles. Concurrently, the activation regions of the small-kernel branch converge toward the centers of individual characteristic peaks to achieve precise localization. Based on the peak intensity fluctuations observed in the spectral iteration, this behavior is likely associated with the fine-tuning of peak heights.

During the late training stage (epochs 40−100), the small-kernel branch transitions toward performing targeted adjustments on fine details. Specifically, it enhances weak peaks within low SNR regions that have not yet been fully restored, such as the minor peak near 2200 cm^{-1} in Figure 3l. Furthermore, it intensifies its focus on regions affected by impulse noise, as evidenced by the noise spikes near 2800 cm^{-1} in Figure 3h and 600 cm^{-1} in Figure 3l. In contrast, the large-kernel branch maintains a sustained focus on continuous peak profiles, a behavior likely aimed at preserving the global structural features of the spectra. At this stage, the model enters a phase of local optimization, with the primary focus being the elimination of noise artifacts and the refinement of local details.

As the training iterations proceed, the Grad-CAM visualizations of the fusion module exhibit an evolution from a dispersed to a focused state. Initially, the activation heatmaps appear relatively scattered; however, they subsequently concentrate and stabilize gradually around significant peaks. By integrating features from the dual-branch encoders, the scSE module assigns higher weights to channels containing valid spectral signals. This function enables the model to precisely localize the critical wavebands associated with feature peaks.

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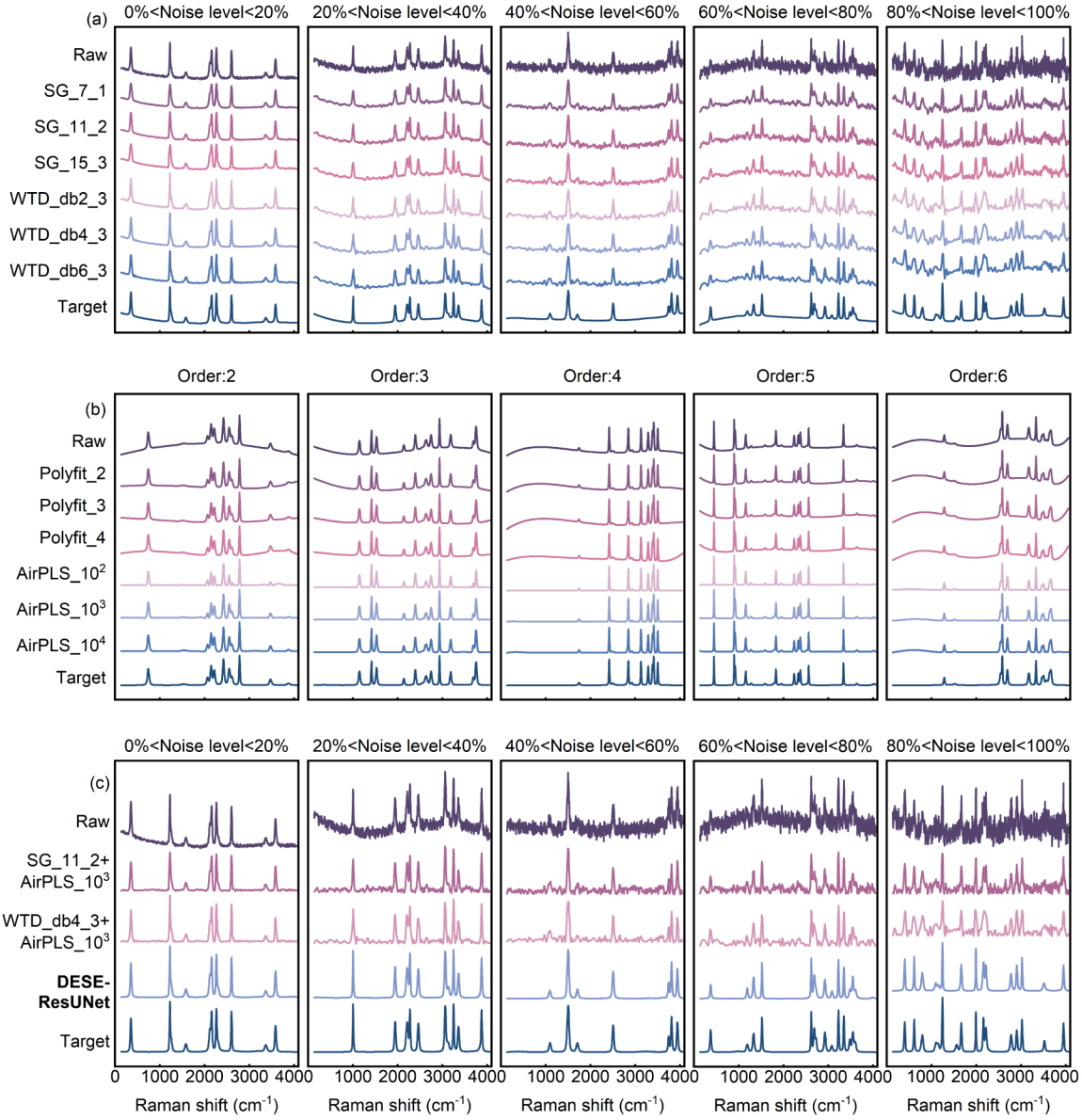


Figure 4. (a) Noise removal performance of SG filter and WTD. (b) Baseline removal performance of polynomial fitting and AirPLS. (c) Baseline correction and noise removal performance of neural network versus traditional algorithms on simulated spectra at 0−100% noise levels.

Performance Comparison between Neural Networks and Traditional Algorithms

The spectral preprocessing performance of the DESE-ResUNet model was evaluated against traditional denoising and baseline correction algorithms using an extra data set comprising 25,000 spectra. For traditional denoising, the Savitzky−Golay (SG) filter and Wavelet Threshold Denoising (WTD) were selected. The specific hyperparameters, including the sliding window size and polynomial order for the SG filter as well as the wavelet basis and decomposition level for WTD, were determined via grid search optimization (Figure S4). The SG filter was configured with window lengths of 7, 11, and 15 and polynomial orders of 1, 2, and 3. For WTD, Daubechies wavelets (db2, db4, and db6) were employed with a decomposition level of 3. Additionally, polynomial fitting and Adaptive Iteratively Reweighted Penalized Least Squares (AirPLS) were selected as the baseline correction algorithms. Under low-noise conditions, spectra are effectively denoised by traditional algorithms. However, in high-noise scenarios, noise cannot be completely eliminated while the details of characteristic peaks are retained (Figure 4a). Among traditional baseline correction algorithms, AirPLS (lambda = 10^3) demonstrates more stable performance (Figure 4b). A further comparison of the complete processing workflow reveals that DESE-ResUNet exhibits significant advantages in spectral preprocessing across low, medium, and high noise levels in example spectra (Figure 4c).

In the comparative preprocessing analysis of the extra data set, spectra processed by DESE-ResUNet exhibited superior performance metrics. Specifically, it outperformed conventional algorithms across average SNR*, PSNR, SSIM, and MRF, with the former three indices showing improvements exceeding 24% (Table 3). Regarding the distribution of performance metrics for individual spectra, DESE-ResUNet consistently outperformed conventional algorithms under increasing noise levels (Figure 5). Although the distributions of SNR*, PSNR, and SSIM for DESE-ResUNet processed spectra exhibited a downward trend as noise intensified, they remained systematically higher than those of traditional methods. It is noteworthy that the decline in

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Table 3. Comparison of Mean Performance Metrics for Preprocessing the Extra Simulated Data Sets between Neural Network and Traditional Algorithms

method	SNR*	PSNR	SSIM	MRF
SG+AirPLS	8.3976	24.3087	0.4839	0.1455
WTD+AirPLS	9.0046	25.4358	0.5465	0.1497
DESE-ResUNet	21.8956	38.4699	0.9402	0.1390

SSIM was characterized by a gradual and steady decrease. The comprehensive results demonstrate that neural network algorithms significantly outperform traditional algorithms in terms of overall performance across different noise levels, including improvements in signal-to-noise ratio, distortion control, and structural preservation.

Effectiveness of the Neural Network on Actual Measured Spectra

To demonstrate the broad applicability of DESE-ResUNet for the reconstruction of experimental Raman spectra, the trained neural network was applied to process empirical data obtained from eight distinct types of microplastics. Spectra exhibiting varying levels of noise interference were acquired by modulating the laser power and integration time, with the total laser energy exposure on the samples being systematically reduced from approximately 780 mJ to 30 mJ. A decrease in received laser energy correlates with intensified spectral noise and degraded spectral quality, thereby posing a more rigorous challenge to the efficacy of the processing algorithms. Using the Raman spectra of PVC microplastics as a representative example, the processing outcomes of the neural network versus traditional algorithms are presented in Figure 6a–h (comparative results for all eight microplastic types are provided in Figure S5). Under low-noise conditions, the DESE-ResUNet model, along with the hybrid traditional methods (SG+AirPLS and WTD+AirPLS), yields comparable results in terms of denoising and baseline correction (Figure 6a–c). However, as noise interference intensifies, traditional algorithms exhibit pronounced residual noise, while the efficacy of the neural network in restoring weak peaks gradually diminishes (Figure 6d–g). Under conditions of severe noise interference where certain Raman characteristic peaks are completely obscured, DESE-ResUNet remains capable of reconstructing prominent signal features. Nevertheless, instances of erroneously identifying noise artifacts as weak peaks begin to emerge, indicating that the model is approaching its performance boundary (Figure 6h). Figure 6i–p illustrate the Grad-CAM heatmaps computed at the final convolutional layer during spectral processing. These results demonstrate that the key wavebands prioritized by the model align remarkably well with the positions of the spectral characteristic peaks.

Figure 7 presents the processing results of DESE-ResUNet applied to Raman spectra of eight distinct microplastics acquired under varying experimental conditions and spectral qualities (Table 4, measured data set 2, 5, 8). When the total laser energy exposure on the sample was reduced to approximately 74 mJ (Table 4, measured data set 5), the neural network successfully reconstructed the primary characteristic peaks. However, the reconstruction efficacy for narrower and weaker peaks was notably compromised, as observed in the PET peak near 600 cm^{-1} (Figure 7c). Even at a significantly lower energy level of approximately 30 mJ (Table 4, measured data set 8), the algorithm retained the capability to reconstruct major characteristic peaks. Nevertheless, under such severe noise interference, minor artifacts emerged, and instances occurred where noise was erroneously identified as weak peaks (Figure 7a). The SNR was calculated as the ratio of the maximum peak intensity to the standard deviation of a nonpeak region (Table 4). Quantitative analysis indicates that the SNR of spectra processed by DESE-ResUNet is 3- to 15-fold higher than that achieved by traditional methods. Overall, DESE-ResUNet effectively performs denoising and baseline correction across diverse noise conditions, substantially enhancing spectral quality.

It must be acknowledged that, due to experimental constraints, all experiments in this study were conducted exclusively using standard samples at a single 785 nm excitation wavelength, a condition that inherently suppresses the fluorescence background. Consequently, further validation is still required to ascertain the applicability of the proposed method using environmentally aged microplastic samples and Raman platforms operating at alternative excitation wavelengths. To demonstrate the broad applicability of the DESE-ResUNet framework for the reconstruction of empirically measured Raman spectra, the trained neural network was

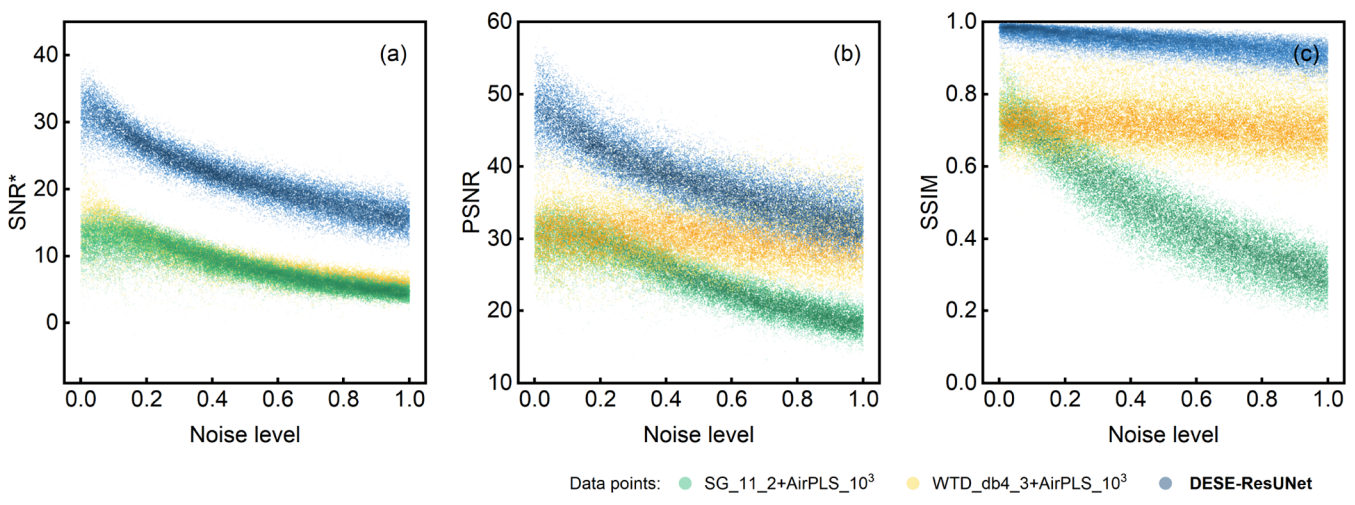


Figure 5. Comparison of neural network versus traditional algorithms for spectral processing on an extra simulated data set. The panels illustrate the distributions of (a) SNR*, (b) PSNR, and (c) SSIM. The horizontal axis represents the noise level, while the vertical axis denotes the corresponding metric values.

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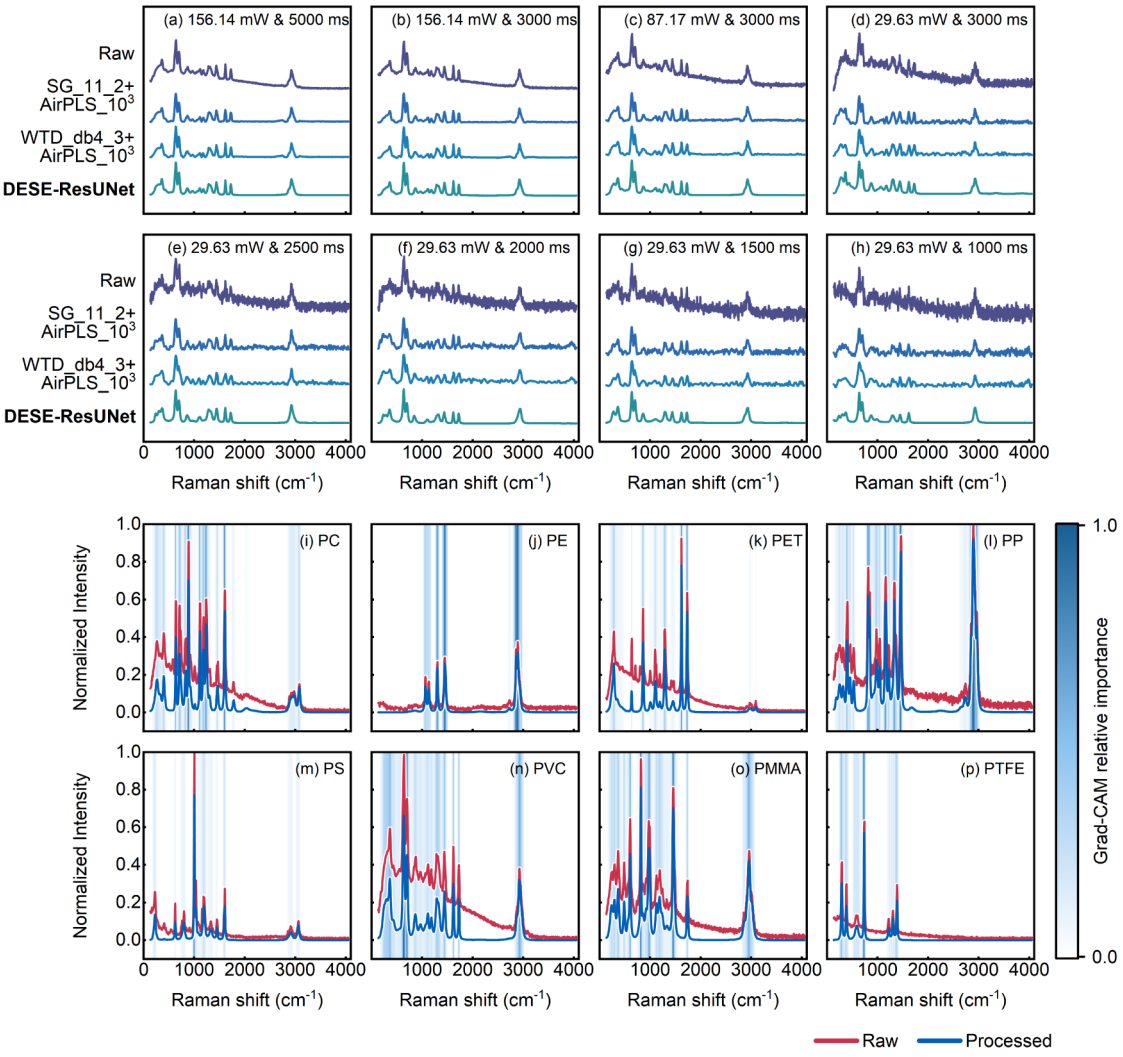


Figure 6. (a–h) Performance comparison between the neural network and traditional algorithms in PVC spectral baseline correction and noise removal under varying experimental noise levels. (i–p) For data collected at a laser power of 156.14 mW and an integration time of 5000 ms, Grad-CAM heatmaps at the final convolutional layer during spectral processing.

utilized to process publicly available Raman spectral data sets (Figure S6a,[51] b,[52] c,[53] d,[54] and e–h[55]). These samples encompass a variety of excitation wavelengths and include diverse sample types, such as microplastics, biological cells, inorganic materials, proteins, and enzymes. Although these public data sets were acquired from distinct experimental platforms and testing conditions, and inherently exhibit significant variations in noise distribution, baseline morphology, and peak width characteristics, the DESE-ResUNet model consistently and stably recovered the clear structural features of the characteristic peaks.

Effect of Spectral Preprocessing on Classification Performance

To evaluate the impact of different data preprocessing methods on classification performance, a ResNet architecture (Figure 1f) and the classic chemometric method, PLS-DA, were employed for spectral classification. Both the classification neural network and the PLS-DA model were trained and constructed using a data set of 2,400 spectra representing eight microplastic types, acquired at a laser power of 156.14 mW and an integration time of 5,000 ms. To prevent the classifiers from developing a bias toward the feature space of a specific preprocessing technique, model training and construction were conducted independently on data sets processed by each respective preprocessing method. Detailed hyperparameter configurations for the classification network are provided in Table S4. The comparative analysis encompassed raw unprocessed spectra, combinatorial approaches pairing either WTD or the SG filter with AirPLS baseline correction, and the proposed DESE-ResUNet method. To mitigate the stochasticity associated with single-run training results, the classification network was trained ten times using controlled random seeds. For the PLS-DA model, construction was performed using 10, 15, and 25 principal components. The detailed cross-classification accuracies of the PLS-DA model and the ResNet model, independently trained on the four preprocessed data sets, are presented in Table S5 (averaged across three principal components configurations) and Table S7 (averaged across ten random seeds), respectively. Furthermore, the overall average classification accuracies for each preprocessing method are summarized in Table S6 for PLS-DA and Table 5 for ResNet. Compared to PLS-DA, the ResNet architecture generally achieved higher classification accuracies, with the exception of models trained on unprocessed spectra, yielding an

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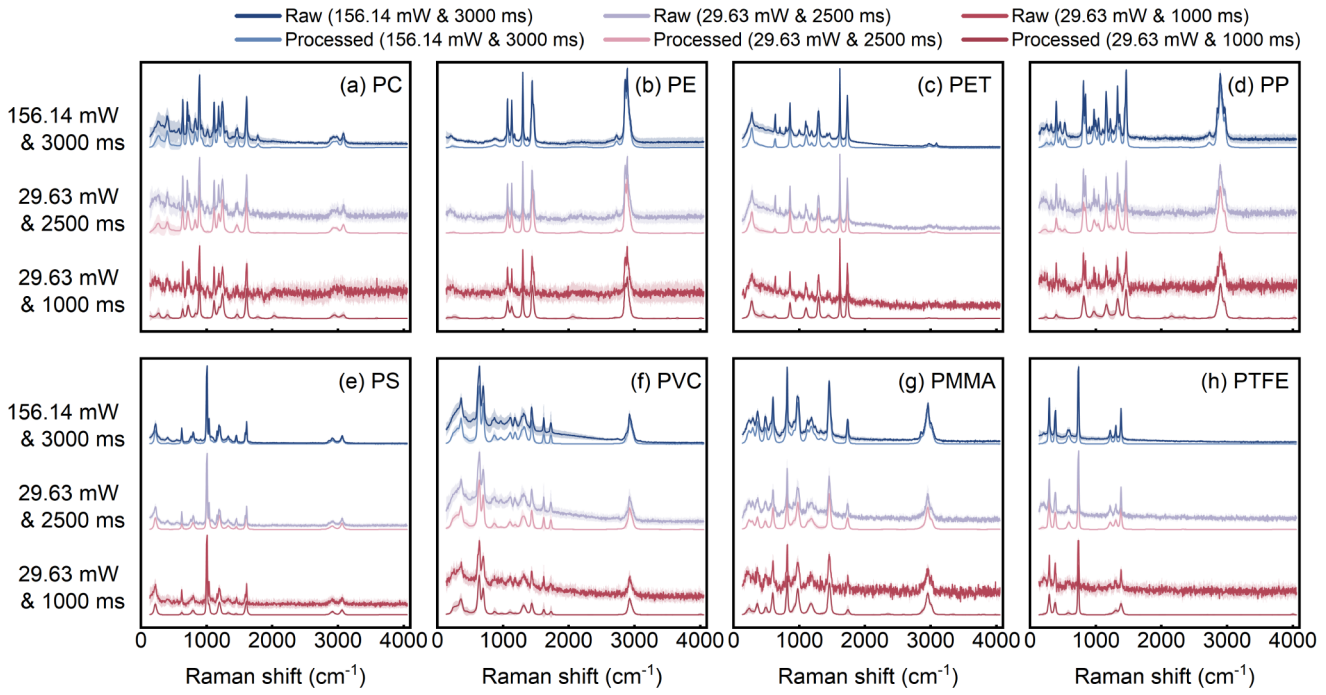


Figure 7. (a–h) Effectiveness of neural network in baseline correction and noise removal for various microplastic Raman spectra under different experimental conditions. Solid lines and shaded areas represent the mean and standard deviation of all collected Raman spectra, respectively.

Table 4. SNR of Different Measured Data Sets Processed by Different Preprocessing Methods

				mean SNR		
measured data set	actual power after fiber (mW)	integration time (ms)	received energy (mJ)	WTD+AirPLS	SG+AirPLS	DESE–ResUNet
1	156.14	5000	780.70	226.36	149.88	**797.01**
2	156.14	3000	468.42	173.56	116.78	**820.39**
3	87.17	3000	261.51	131.67	88.69	**693.98**
4	29.63	3000	88.89	69.87	44.93	**480.18**
5	29.63	2500	74.08	59.38	39.03	**453.21**
6	29.63	2000	59.26	52.25	33.14	**422.81**
7	29.63	1500	44.45	36.47	24.19	**441.21**
8	29.63	1000	29.63	25.60	18.35	**423.89**

Table 5. Average Classification Performance of Neural Networks Trained with Various Preprocessed Spectra across Different Experimental Conditions and Preprocessing Methods

					mean accuracy after data processing (%)			
measured data set[a]	nominal electric current (%)	actual power after fiber (mW)	integration time (ms)	energy received (mJ)	raw	WTD+AirPLS	SG+AirPLS	DESE–ResUNet
2	75	156.14	3000	468.42	92.19	92.82	97.91	**99.26**
3	65	87.17	3000	261.51	87.80	91.49	97.56	**99.23**
4	55	29.63	3000	88.89	72.08	86.56	91.09	**98.04**
5	55	29.63	2500	74.08	69.56	84.99	85.92	**95.62**
6	55	29.63	2000	59.26	61.53	83.07	83.68	**94.41**
7	55	29.63	1500	44.45	51.68	77.30	69.73	**94.01**
8	55	29.63	1000	29.63	41.54	66.83	55.62	**82.80**

[a]The measured data sets presented in this table are identical to the actually measured data sets sharing the same numbering in Table 4.

average performance improvement of 14.22% across the data sets.

As indicated by the classification results of the ResNet neural networks trained independently on the differently preprocessed data sets (Table S7), performance generally improves when the preprocessing methods applied to the training and testing sets are identical. This enhancement is attributed to the classifier learning the specific feature space distribution patterns of the corresponding preprocessed data. This phenomenon is particularly pronounced in low-quality spectra (Table 5, measured data sets 7 and 8). For instance, models trained on raw data yield the optimal classification results for raw data. Similarly, classifiers trained on data processed via the combinatorial SG and AirPLS approach achieve higher accuracies on identically processed data than on data subjected to other preprocessing methods. Nevertheless, based on the

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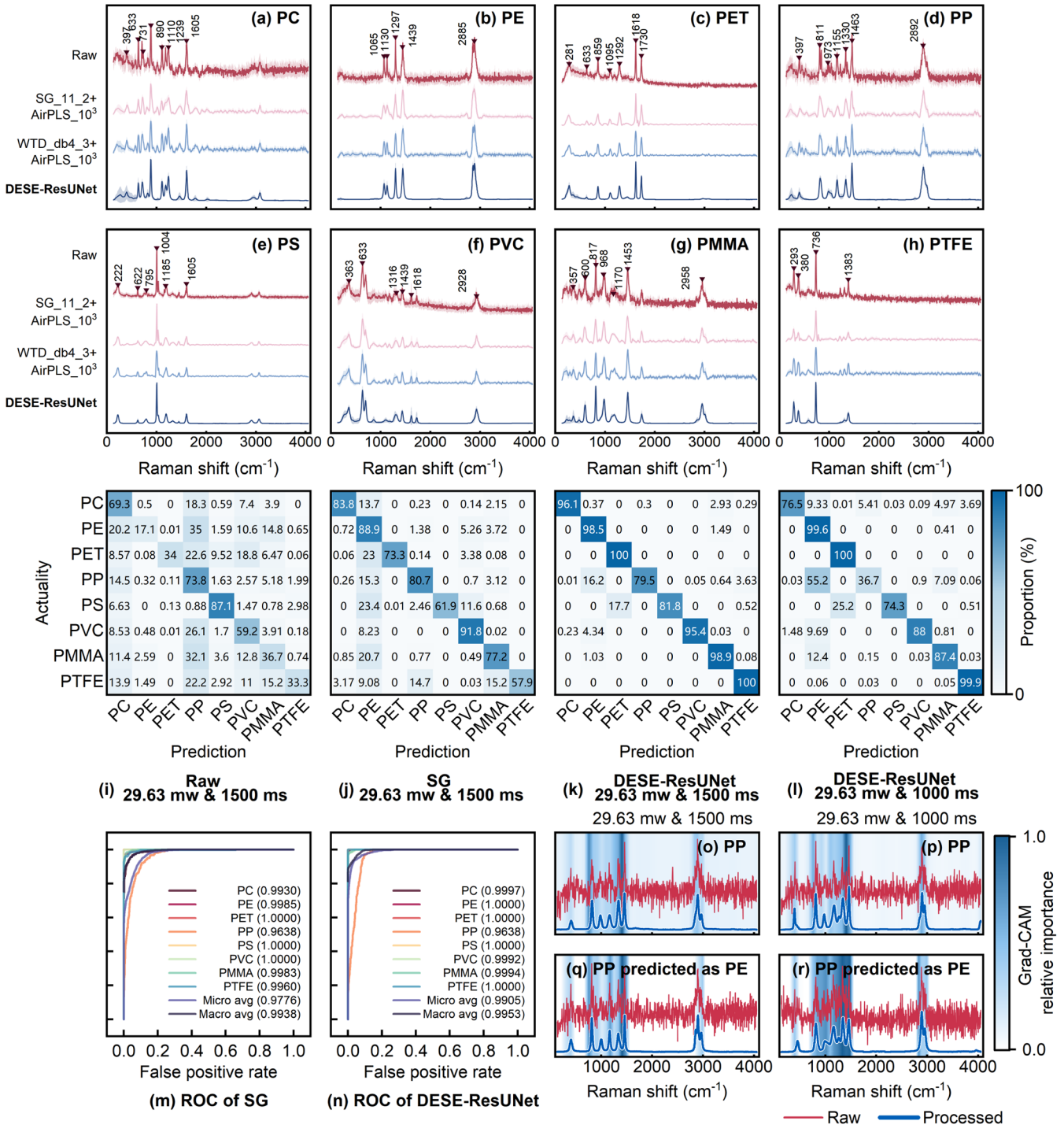


Figure 8. Classification performance of neural network after different data preprocessing methods. (a−h) Spectral comparisons under 29.63 mW and 1500 ms for different data processing methods. Solid lines represent spectral averages, while shaded areas indicate standard deviations. (i−l) Confusion matrices for neural network classification. (m, n) ROC curves comparing classifications based on SG+AirPLS-processed and DESE-ResUNet-processed spectra at 29.63 mW laser power and 1000 ms integration time. (o−r) Correctly and incorrectly classified PP spectra alongside Grad-CAM heatmaps.

overall average classification accuracies for each preprocessing method evaluated across the four model configurations (Table 5), the highest average accuracy is consistently achieved when models are evaluated against test sets processed by DESE-ResUNet, irrespective of the preprocessing data used during training. This demonstrates that DESE-ResUNet preserves classification-relevant spectral features more effectively, thereby enabling classifiers to achieve highly stable and precise recognition performance even without being specifically trained on its specific feature space.

Further analysis of the classification performance under varying SNR conditions reveals that when spectral quality is high, unprocessed spectra still yield relatively high recognition accuracy. However, spectral preprocessing via denoising and baseline correction significantly enhances classification performance (Table 5, measured data sets 2 and 3). As spectral quality

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Table 6. Comparison of Microplastic Classification Performance Metrics under Different Experimental Conditions (30 mW, 1000/1500 ms)

		30 mW 1500 ms			30 mW 1000 ms		
method	category	precision	recall	f1-score	precision	recall	f1-score
raw	PC	0.6359	0.7075	0.5824	0.5802	0.5943	0.4603
	PE	0.3703	0.1713	0.2135	0.4184	0.1146	0.1618
	PET	0.6466	0.3399	0.3779	0.4275	0.2714	0.2870
	PP	0.5023	0.7353	0.5066	0.3266	0.6827	0.3692
	PS	0.8886	0.8464	0.8354	0.8661	0.7246	0.7404
	PVC	0.5627	0.6109	0.4852	0.4902	0.5036	0.4045
	PMMA	0.4656	0.4143	0.3507	0.3724	0.2691	0.2240
	PTFE	0.4550	0.3295	0.3442	0.4968	0.1859	0.2282
	macro avg	0.5691	0.5168	0.4635	0.5001	0.4154	0.3608
	accuracy			0.5168			0.4154
SG+AirPLS	PC	0.9650	0.8464	0.8590	0.9032	0.7266	0.7577
	PE	0.7767	0.8874	0.7514	0.7646	0.8233	0.6954
	PET	0.8998	0.7335	0.7595	0.8997	0.6382	0.6900
	PP	0.8855	0.8199	0.7965	0.7954	0.6074	0.5996
	PS	0.6982	0.6187	0.6349	0.6630	0.5993	0.6132
	PVC	0.8510	0.9177	0.8571	0.7702	0.8994	0.7846
	PMMA	0.7952	0.7719	0.7062	0.5787	0.7005	0.5446
	PTFE	0.9500	0.5884	0.6799	0.9500	0.3524	0.4525
	macro avg	0.8562	0.7730	0.7577	0.7958	0.6683	0.6459
	accuracy			0.7730			0.6683
DESE-ResUNet	PC	0.9977	0.9650	0.9799	0.9803	0.7648	0.8562
	PE	0.8714	0.9851	0.9147	0.5905	0.9959	0.7269
	PET	0.8746	1.0000	0.9271	0.8314	1.0000	0.8996
	PP	1.0000	0.8090	0.8620	0.8921	0.3669	0.4784
	PS	1.0000	0.8180	0.8805	0.9999	0.7412	0.8220
	PVC	0.9994	0.9549	0.9747	0.9880	0.8819	0.9216
	PMMA	0.9570	0.9889	0.9714	0.8910	0.8729	0.8647
	PTFE	0.9625	1.0000	0.9796	0.9603	0.9987	0.9788
	macro avg	0.9578	0.9401	0.9362	0.8919	0.8280	0.8187
	accuracy			0.9401			0.8280

progressively deteriorates, the classification accuracy for unprocessed spectra declines; in contrast, spectra processed with the preprocessing algorithms sustain stable and high recognition accuracy. When the sample laser energy exposure was approximately 44.45 mJ (Table 5, measured data set 7), the classification accuracy for spectra processed by traditional algorithms declined to below 80%, whereas DESE-ResUNet maintained an accuracy exceeding 94%. Figure 8a–h illustrates the spectral comparison of the different data processing methods under these experimental conditions. Meanwhile, Figure 8i–l presents the average confusion matrices, derived from ten training runs, for the models trained on differently preprocessed data across the three processing methods. In the absence of spectral processing, PS exhibits high recognition accuracy, attributed to its distinct characteristic peak at 999 cm^{-1}. The Raman spectra of the other microplastics are compromised by varying degrees of baseline drift in the low-wavenumber region, alongside noise interference. Following preprocessing via the SG filter and AirPLS, the classification accuracies for PC, PET, PS, PVC, and PMMA microplastics improved significantly.

However, instances of spectral misclassification still occur, such as the misidentification of PE and PP. The misclassification between PE and PP is primarily due to their adjacent characteristic spectral bands in the 1000–1500 cm^{-1} region. For PE, these features are mainly attributed to the symmetric and antisymmetric C–C stretching vibrations (around 1060–1120 cm^{-1}) and the CH_2 symmetric deformation and scissor vibrations (around 1430–1450 cm^{-1}). In PP, the characteristic peaks in this region are mainly associated with C–C backbone stretching coupled with CH_3 rocking and C–CH_3 bending modes (around 1149–1160 cm^{-1}), as well as CH_3 asymmetric bending and CH_2 bending (around 1448 cm^{-1}). Under low signal-to-noise ratio conditions, the peak shapes of these vibrational modes are relatively broad with limited spectral resolution, potentially exhibiting similar spectral features and thus increasing the likelihood of confusion. Similarly, the misclassification of PTFE spectra as PVC may be related to their similar waveforms in the 600–800 cm^{-1} range. In PTFE, the characteristic peaks in this region are mainly related to CF_2 stretching and backbone vibrations, whereas in PVC, they originate from CH rocking modes and both crystalline and noncrystalline C–Cl stretching.[56] In the presence of residual noise, these partially overlapping spectral features cannot be correctly classified by the neural network.

Furthermore, following processing with the neural network, the average F1-score for the various microplastics increased from 0.4635 to 0.9362, significantly outperforming the combinatorial SG and AirPLS approach, which achieved a score of 0.7577 (Table 6). For one of the classification rounds, the concentration of the ROC curves toward the top-left corner and the correspondingly higher AUC values further demonstrate that the dual neural network combination yields superior results (Figure 8m, n). Even under the least ideal experimental conditions evaluated (Table 5, measured data set 8), the

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DESE-ResUNet model maintained an average classification accuracy of approximately 82.80%, whereas traditional methods achieved only 66.83%. The classification confusion matrix indicates that the primary error involves the misidentification of PP as PE, likely because the positions of their characteristic peaks are in close proximity within the 1000−1500 cm^{-1} range. Figure 8o−r displays representative PP spectra, including both correctly classified samples and those misclassified as PE, alongside their Grad-CAM heatmaps derived from the last convolutional layer of the classification network. When PP is misclassified as PE, the model exhibits heightened attention to the wavenumber region near 1100 cm^{-1}. Consequently, if the preprocessing method fails to accurately restore the spectral peak at this position, the classification network is more prone to recognition errors. Additionally, Figure S7 presents a comparison of the spectra processed using different methods under the most challenging experimental conditions. Representative spectra and Grad-CAM heatmaps for all correctly classified microplastic types are presented in Figure S8.

However, it must be noted that the robustness evaluation of the model in this study relied primarily on establishing varying signal-to-noise ratio conditions by reducing laser energy. The system has not yet been systematically evaluated against the complex perturbations likely encountered during actual field measurements.[57,58] Operating in authentic in situ environments inevitably subjects Raman spectra to interference from multiple non ideal factors. These encompass thermal detector drift caused by the thermal expansion and contraction of optical components due to ambient temperature fluctuations, wavelength shifts and vibration artifacts induced by mechanical instability, and complex fluorescence background fluctuations arising from natural organic matter within cocontaminants in the environmental matrix. Moreover, experiencing natural environmental aging processes frequently subjects microplastics to reduced crystallinity and molecular chain scission, or oxidation reactions that produce new oxygen-containing functional groups. These physical and chemical alterations subsequently cause the characteristic Raman peaks to exhibit varying degrees of broadening, deformation, or shifting, alongside the potential generation of anomalous interference peaks completely absent from the standard spectral profiles. The current neural network model relies heavily on supervised training utilizing standard microplastic spectra acquired under tightly controlled conditions. While this approach has demonstrated exceptional performance in removing shot noise, reconstructing characteristic peaks, and improving classification accuracy throughout the conducted experiments, its broader applicability across highly complex field scenarios remains inherently constrained by the representativeness of the training data. Future efforts to comprehensively evaluate and further enhance the generalization capability of the model within complex in situ environments will necessitate the implementation of highly standardized perturbation simulation experiments. Executing these targeted trials is essential to effectively expand the training data set with diverse and representative real-world scenarios.

CONCLUSIONS

In this work, the DESE-ResUNet model, constructed with a dual-branch encoder and a feature fusion module, is proposed for Raman spectral denoising and baseline correction. Through quantitative evaluation based on the NSQE metric, it was confirmed that the simulated data set incorporating measured instrumental noise exhibits the highest spectral feature similarity to real spectra, thereby establishing it as the optimal training data. Furthermore, Grad-CAM was utilized to elucidate the decision-making basis of the dual-branch encoder during the denoising and baseline correction processes, revealing the distinctions and correlations in their feature extraction mechanisms. Processing results derived from simulated data demonstrated that Raman spectral quality under varying noise levels was effectively improved by the neural network model. Enhancements exceeding 24% in key metrics such as SNR improvement, PSNR, and SSIM were achieved compared to traditional SG filter and WTD algorithms. Regarding the processing of experimental microplastic Raman spectra, the signal-to-noise ratio of spectra processed by DESE-ResUNet was found to be 3- to 15-fold higher than that obtained via traditional methods. Comprehensive capabilities regarding SNR improvement, distortion control, structure preservation, and fluctuation control were demonstrated across different noise levels. Under the least ideal experimental conditions evaluated, the classification recognition accuracy of spectra processed by the DESE-ResUNet neural network was elevated from an unprocessed baseline of 41.54% to 82.80%, whereas traditional methods achieved only 66.83%. Future work will integrate controlled environmental experiments to construct data sets encompassing the aforementioned field perturbation factors, aiming to systematically evaluate and enhance the generalization capability of the model in complex practical application scenarios. Designing a more stable and efficient network architecture, alongside the optimization of model training strategies, will also constitute an important direction for future research.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.analchem.6c02067.

Detailed method of the Natural Spectra Quality Evaluator and other evaluation metrics, code operational environment dependencies, neural network hyperparameter settings for spectra reconstruction and classification tasks, Grad-CAM visualizations of important features during training, performance benchmarking of traditional denoising algorithms, and more results of microplastic Raman spectra baseline correction, noise reduction, and classification (PDF)

Representative spectral output results and importance feature visualization after each training epoch during network training (MP4)

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Notes

The authors declare no competing financial interest.

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